

Program 5: CI Integration using Jenkins in Docker

Step 1: Check prerequisites & create project folder

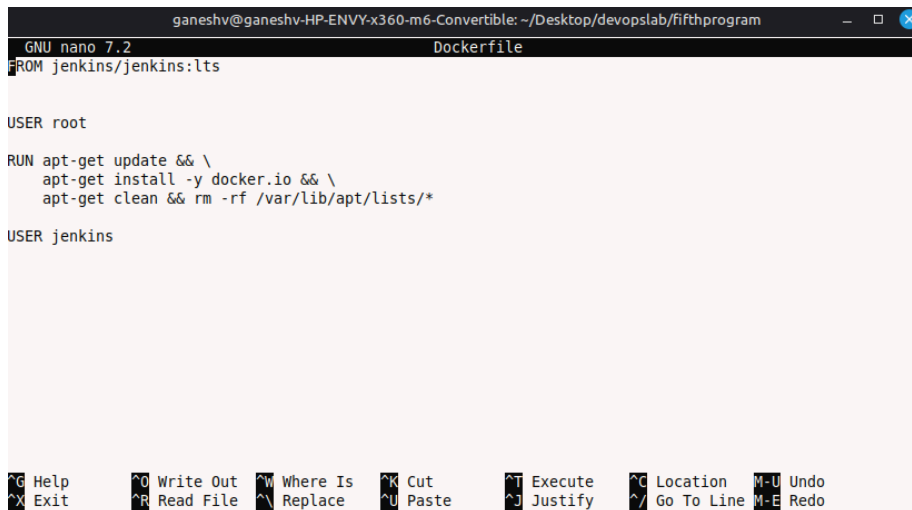
```
docker --version
docker compose version
mkdir program5
cd program5
```

Step 2: Create Dockerfile

```
nano Dockerfile
```

```
FROM
jenkins/jenkins:lts USER
root
RUN apt-get update && apt-get install -y docker.io
USER jenkins
```

Save & exit.

A screenshot of a terminal window titled "ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/fifthprogram". The terminal shows the nano 7.2 editor editing a file named "Dockerfile". The content of the Dockerfile is as follows:

```
FROM jenkins/jenkins:lts

USER root

RUN apt-get update && \
    apt-get install -y docker.io && \
    apt-get clean && rm -rf /var/lib/apt/lists/*

USER jenkins
```

The terminal window has a standard Linux desktop environment with window controls and a status bar at the bottom showing various keyboard shortcuts like Help, Exit, Write Out, Read File, etc.

Step 3: Create docker-compose.yml

```
nano docker-compose.yml
version: '3.8'
```

```
services:
  jenkins:
    build: .
```

container_name: jenkins-ci

ports:

- "8080:8080"
- "50000:50000"

volumes:

- jenkins_home:/var/jenkins_home
- /var/run/docker.sock:/var/run/docker.sock

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/fifthprogram
GNU nano 7.2 docker-compose.yml
version: '3.8'

services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock
    restart: unless-stopped

volumes:
  jenkins_home:
```

volumes:

jenkins_home:

Step 4: Build and run Jenkins

docker compose up --build

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/fifthprogram
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/fifthprogram$ nano Dockerfile
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/fifthprogram$ nano docker-compose.yml
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/fifthprogram$ docker compose up --build
WARN[0000] /home/ganeshv/Desktop/devopslab/fifthprogram/docker-compose.yml: the attribute `version` is ob
solete, it will be ignored, please remove it to avoid potential confusion
[+] Building 2.5s (9/9) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 5558 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 202B 0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts 2.2s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c15 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io && apt-get clean && rm 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:fef175da3330946ae2b49667e0bd3d4e8ad5bfc496fc097a66e0b23ce9936695 0.0s
=> => naming to docker.io/library/fifthprogram-jenkins 0.0s
=> resolving provenance for metadata file 0.0s
[+] Running 2/2
✓ fifthprogram-jenkins Built 0.0s
✓ Container jenkins-ci Running 0.0s
Attaching to jenkins-ci
```

Step 5: Access Jenkins UI

Open browser:

<http://localhost:8080>

